

Scientific Paper Domain Classification: Comparing TF-IDF, GloVe Recurrent Networks, and BERT

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Abstract

The rate at which scientific information is increasing in various research areas is growing fast, and the automatic organization and classification of research papers is becoming increasingly useful in digital libraries, academic search systems, recommendation systems and literature management. The purpose of this study is to automatically classify scientific paper abstracts into seven scientific parent domains, namely Computer Science, Electrical Engineering, Psychology, Mechanical Engineering, Civil Engineering, Medical Science and Biochemistry. We consider a dataset of 11,967 scientific documents (WOS-11967), and we compare three families of text representations and classifiers. Logistic Regression, Multinomial Naive Bayes and Random Forest are used for a traditional machine learning approach using TF-IDF representations. 100-dimensional GloVe embeddings, SimpleRNN, GRU, LSTM, and their bidirectional versions are used in neural sequence models. Finally, a bert-base-uncased transformer is fine-tuned with WordPiece representations. The data is split into three datasets with a fixed ratio of 70/15/15. The model selection is done by using the validation macro-F1 score, and a final evaluation is conducted on the held-out test set. BERT gets best macro-F1 score among the ten models evaluated, with SimpleRNN getting the lowest score 0.4438. The final BERT model has a test accuracy of 95.16% and a macro-F1 of 95.06%. The best recurrent models are bidirectional GRU (macro-F1 = 90.56%) and the best traditional classifiers are Logistic Regression (macro-F1 = 90.27%). The results provide evidence of the benefit of contextual transformer representations for scientific-domain classification, and that a carefully configured TF-IDF and recurrent approach are competitive at significantly less computational cost.

Learning objectives. The student must understand the basic concepts of scientific text classification, document classification, TF-IDF, GloVe

model, RNN, GRU, LSTM, BERT, scientific abstracts and NLP.

1 Introduction

As the number of scientific publications has been increasing rapidly, there is a big information management problem. Large collections of scientific documents are increasingly in need of being organized by researchers, students, librarians, and information-retrieval systems by their research areas. Manual classification is costly and labor-intensive, while automatic classification can be a cost-effective means of assigning documents to predefined scientific fields of classification. These are well suited for this purpose because scientific abstracts have a short description of the research problem, research method, research results, and the application area.

The classification of science domains is more difficult than the classification of any other topic as many different research domains use the same terminology. For instance, in an engineering paper, the medical or biological terms might be used if a biomedical application is being discussed, and in a computer science paper, the medical or biological terms could be used if the human-computer interaction is being discussed. Thus, to be effective, a classifier should be able to recognize words used in the specific domain as well as understand the relationships between the words in context.

This project explores classification of scientific paper abstracts from WOS-11967 data set to seven parent domains. The data set consists of 11,967 documents and comprises seven parent-level categories: Computer Science, Electrical Engineering, Psychology, Mechanical Engineering, Civil Engineering, Medical Science and Biochemistry. The data is from the dataset created by [Kowsari et al. \(2018\)](#) and published by the Web of Science collection.

The project is a comparison of pretrained word embeddings, traditional statistical text representations and contextual transformer representations. TF-IDF gives a sparse representation of the word or phrase importance, and GloVe gives dense word vectors pre-trained. BERT gives us contextually token representations that can be used to represent the meaning of a word based on its context.

The experiments are guided by four research questions:

- RQ1: What is the best TF-IDF traditional classifier for classification in scientific domain?
- RQ2: What is the impact of recurrent cell type and bidirectionality on macro-F1?
- RQ3: Is BERT-base cost-effective over the best non-BERT baselines?
- RQ4: What are the most common misconceptions for scientific domains and what are some of the characteristics that could lead to these?

Experimental design is focused on repeatability and leakage. Prior to model fitting, the data are split into training, validation and test partitions. Normalized text is used to duplicate documents and duplicate groups are disallowed to span across dataset partitions. The model configurations are chosen based on the validation macro-F1, and the test set is kept isolated until final evaluation.

The rest of the paper is organized as follows. Section 2 reviews relevant previous work and the representations and models used in this study. In Section 3, the data, exploratory analysis, data preprocessing, representations and model architectures are introduced. The results of the experiment and error analysis are presented in Section 4. The paper ends with limitations and directions in Section 5.

2 Related Work

Traditionally, automatic text classification has been done using manually designed or statistically derived text representations and supervised machine-learning algorithms. One of the most popular representations is the TF-IDF representation, which assigns more weight to terms that are important in a document but fewer to those that are common across the documents in a corpus. Deep learning models with sparse lexical representations and linear classifiers have proven to be good baselines for document classification.

Another key strand in NLP research was neural word representations. Mikolov et al. (2013) showed that distributed word representations can embody useful semantic and syntactic relationships between words. Another popular solution, GloVe, was introduced for learning word representations globally from a large text corpus (Pennington et al., 2014). Unlike TF-IDF, GloVe is stored as a dense vector and neural models can leverage the semantic relationship between words.

The sequential relationships in text were then modeled using recurrent neural networks. Conventional recurrent architectures are not well suited to retaining information over long sequences, while the standard RNNs process tokens sequentially. In response to the vanishing-gradient problem, the memory cells and gating mechanisms were presented by Hochreiter and Schmidhuber (1997). While maintaining the capability to capture longer dependencies, GRU offers a simpler gated recurrent architecture (Cho et al., 2014). Bidirectional recurrent networks can be used to process sequences in both directions, with a representation being able to use information from both sides of a token.

Representation learning of NLP has significantly evolved with the transformer models. Vaswani et al. (2017) proposed the Transformer architecture that is based on self-attention instead of recurrence. BERT also showed the usefulness of pretrained bidirectional transformer representations for language understanding problems (Devlin et al., 2019). Instead of a one-to-one correspondence between word and representation, BERT produces a representation that is contextually dependent and thus varies based on the context of the sequence it is applied to.

Hierarchical and neural methods are also explored in scientific text classification. The Web of Science collection, which is the basis of the dataset used in this project, was used by Kowsari et al. (2017) to introduce a hierarchical deep-learning framework for text classification. The WOS dataset is hierarchical and consists of parent-level and more fine-grained child-level categories suitable for hierarchical scientific document classification.

The present work is an elaboration on the single-model classification study and a controlled comparison is made between three representation families: TF-IDF, GloVe, and BERT WordPiece. It also compares several recurrent architectures and explicitly investigates whether or not bidirectionality helps. In all the experiments, the same train/test split and

Domain	Documents	Percentage
Computer Science	1,499	12.53%
Electrical Engineering	1,132	9.46%
Psychology	1,959	16.37%
Mechanical Engineering	1,925	16.09%
Civil Engineering	2,107	17.61%
Medical Science	1,617	13.51%
Biochemistry	1,728	14.44%
Total	11,967	100%

Table 1: Dataset class distribution.

the same macro-F1-based model selection method is employed to keep the model families under a common evaluation framework.

3 Methodology

3.1 Dataset

The experiments are based on version 6 of the Web of Science dataset: WOS-11967. Data is made available by Mendeley Data (<https://doi.org/10.17632/9rw3vkcfy4.6>), is licensed under CC BY 4.0 and is made available under the same license. This dataset was compiled by Kamran Kowsari, Donald Brown, Mojtaba Heidarysafa, Kiana Jafari Meimandi, Matthew Gerber, and Laura Barnes (Kowsari et al., 2018).

The project takes $X.txt$ as its document collection for learning and $YL1.txt$ as its target variable. The child-level labels in $Y.txt$ and $YL2.txt$ were excluded. The seven classes are Computer Science, Electrical Engineering, Psychology, Mechanical Engineering, Civil Engineering, Medical Science, and Biochemistry.

The entire collection consists of 11,967 abstracts. The class distribution is given in Table 1.

Civil Engineering has the highest number of documents with 2,107 documents, with Electrical Engineering having the least with 1,132 documents. The class size ratio is not extremely skewed (highest to lowest class size is 1.86). This imbalance makes it appropriate to use macro-F1 as the major model-selection metric.

The average number of words used in the abstracts is 195.17 with a median of 189 words and a standard deviation of 70.81 words. The shortest abstract is 22 words long and the longest is 906 words. The 95th percentile is 319 words and the 99th percentile is 407 words. The number of characters of an average abstract is around 1,340.77.

The domain-specific length statistics also show meaningful variation. Medical Science has the

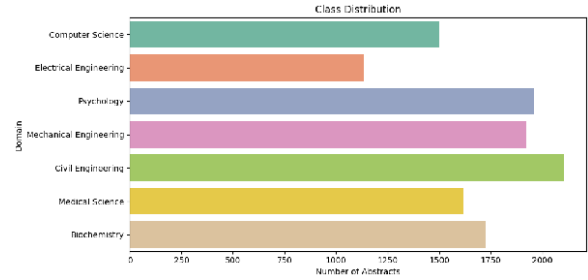


Figure 1: Distribution of the seven scientific parent domains.

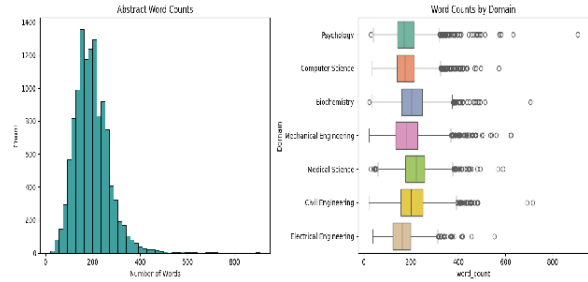


Figure 2: Distribution of abstract word counts and word counts by scientific domain.

highest average abstract length at 221.68 words, followed by Biochemistry at 208.58 and Civil Engineering at 208.07. Electrical Engineering has the shortest average abstracts at 165.34 words.

Word clouds were generated separately for all seven domains. These visualizations demonstrate that the classes contain recognizable domain-specific terminology, although several technical words occur across multiple disciplines.

The quality checks did not identify any missing abstracts, empty abstracts, missing labels, exact duplicate rows or normalized duplicate rows. There were no conflicting labels between duplicate groups. These checks help to minimise the risk of artificially inflated model performance due to duplicated documents or inconsistent labels.

3.2 Data Splitting and Leakage Prevention

For all experiments a fixed train-validation-test split was employed. The final partitions have 8,376 training, 1,795 validation, and 1,796 test examples, roughly 70/15/15.

The split was made prior to representation fitting. Quality checks found no exact or normalized duplicate abstracts. Therefore, a standard stratified 70/15/15 row-level split was used without the risk of duplicate abstracts appearing across partitions. The split manifest and leakage assertions were stored in the executed dataset, which did not



Figure 3: Word clouds generated separately for the seven scientific domains.

contain any normalized duplicates.

The test set was not involved in the process of hyperparameter selection. Each one of the ten required models was tested with three configurations, for a total of 30 official validation runs. Each model was optimised with the help of the validation macro-F1, and then accuracy was used as a tie-breaker. The test set was then used after making these selections.

This procedure also helps to avoid the selection of a model based on test performance and gives a purer estimate of generalization.

3.3 Preprocessing

There are different approaches to preprocessing for different representations. A major advantage of the project is therefore that there will be no universal preprocessing of models.

In TF-IDF, text is normalized to lowercase and Unicode normalization and whitespace normalization are performed. This is because there is information that can be contained within phrases that cannot be conveyed by a single word, like multi-word technical concepts in scientific phrases.

In case of the recurrent models (GloVe), the tokenizer is fitted only to the abstracts in the training set. The vocabulary includes only the 30,000 most frequent words, and an explicit out-of-vocabulary

token (OOV). The length of the sequence depends on the training distribution. A maximum length of 320 tokens is selected for a training sequence length of 95th percentile. Sequences are padded and truncated at their end.

The official tokenizer provided with bert-base-uncased is used for BERT. The BERT tokenizer is a bit different from the TF-IDF preprocessing pipeline in that it will do WordPiece tokenization and will not need much manual normalization. The median training token length was found to be 254 tokens, the 95th percentile was 473.25 tokens and the maximum length was 1,119 tokens. Due to the fact that 48.85% of the training examples had a length of more than 256 tokens, the maximum BERT sequence length was determined to be 384.

One of the limitations is the long-length truncation of documents in the selected sequence length, meaning information after the chosen sequence length will be lost.

3.4 Word Representations

3.4.1 TF-IDF

TF-IDF is the first representation. The vectorizer is based on lowercased text, normalized preprocessing, unigram and bigram features, maximum document frequency of 0.95, minimum document frequency of 2 and sublinear scaling.

The vectorizer is only trained on the training documents. The training representation has 8,376 documents and 50,000 features.

TF-IDF gives an interpretable lexical baseline where each dimension is based on a term or short phrase. It does, however, not explicitly model contextual meanings. The same word is represented in the same way, no matter whether it is in one sentence or another one.

3.4.2 GloVe

The second representation is based on pretrained 100-dimensional GloVe vectors from GloVe 6B (Pennington et al., 2014). If a word in the training vocabulary is encountered, its corresponding embedding vector is initialized using the pretrained vector. Random initialization is used for unmatched words.

The final training vocabulary includes 30,000 words. The selected vocabulary is approximately 83.23% covered with GloVe and the out-of-vocabulary rate of validation tokens is approximately 3.15%.

Model	Config.	Main hyperparameters
Logistic Regression	LR_1	$C = 0.5$
Logistic Regression	LR_2	$C = 1.0$
Logistic Regression	LR_3	$C = 2.0$
Multinomial NB	NB_1	$\alpha = 0.1$
Multinomial NB	NB_2	$\alpha = 0.5$
Multinomial NB	NB_3	$\alpha = 1.0$
Random Forest	RF_1	200 trees, depth 50
Random Forest	RF_2	300 trees, depth 100
Random Forest	RF_3	400 trees, unlimited

Table 2: Traditional model configurations.

The recurrent models share the same representation, that is, GloVe. In some experiments the embedding layer is frozen and in profile C the embeddings are trainable.

3.4.3 BERT WordPiece

The third representation is produced by the pre-trained bert-base-uncased model (Devlin et al., 2019). BERT features WordPiece tokenization and contextual transformer representations. BERT representations are dynamic, as opposed to static GloVe vectors, which rely on the words that are in their context.

There are seven output labels in the BERT classifier representing the seven scientific domains. Fine-tuning is performed with three learning rates: 1×10^{-5} , 2×10^{-5} , and 3×10^{-5} .

3.5 Model Architectures

3.5.1 Traditional Models

The same TF-IDF representation is used for three traditional classifiers.

C values of 0.5, 1.0, and 2.0 are used to test Logistic Regression. There is a maximum of 1,000 iterations.

We test Multinomial Naive Bayes using smoothing parameters $\alpha = 0.1, 0.5$, and 1.0.

Random Forest is tested with three configurations: 200 trees with maximum depth 50; 300 trees with maximum depth 100; and 400 trees with unlimited depth.

3.5.2 Recurrent Neural Models

Six recurrent architectures are evaluated: SimpleRNN, GRU, LSTM, Bidirectional SimpleRNN, Bidirectional GRU, and Bidirectional LSTM.

The 100-dimensional GloVe embedding layer is shared amongst all the recurrent models, with the recurrent layer, dropout layer, and a seven-class softmax output layer thereafter. Adam and sparse

Profile	Units	Train.	Drop.	LR
A	64	No	0.3	0.001
B	128	No	0.3	0.001
C	128	Yes	0.5	0.0005

Table 3: Recurrent model profiles. “Train.” indicates whether GloVe embeddings are trainable.

categorical cross-entropy are used to optimize the models.

All three profiles are compared for each of the recurrent architectures.

The training is done for a maximum of 15 epochs with a batch size of 64. Early stopping monitored validation loss with a patience of two epochs, while the checkpoint with the lowest validation loss was retained.

3.5.3 BERT

The final model family is BERT-base-uncased. It is a combination of a pretrained BERT encoder and a seven-class sequence-classification head. The model is trained for three epochs with a batch size of 8 and weight decay of 0.01.

Three learning rates are compared: 1×10^{-5} , 2×10^{-5} , and 3×10^{-5} . The best configuration in terms of the highest validation macro-F1 score is chosen.

4 Results

4.1 Validation-Based Model Selection

The ten models required were tested with three configurations, giving 30 formal runs. A validation macro-F1 was used to find the best configuration for each model.

The validation results show already a significant gap between the model families. The best validation macro-F1 score is given by BERT, followed by Logistic Regression, Random Forest, and Bidirectional GRU.

4.2 Final Test Results

Once each model was configured, each model was then tested on the test set that it had never seen before.

BERT obviously delivers the optimal overall performance. It has a macro-F1 score of 95.06%, which is about 4.50 percentage points higher than its closest competitor, the non-BERT model Bidirectional GRU.

Model	Representation	Selected configuration	Val. Acc.	Val. Macro-F1
BERT Base	BERT WordPiece	BERT_3	94.60%	94.55%
Bidirectional GRU	GloVe 6B 100d	B	89.92%	89.96%
Bidirectional LSTM	GloVe 6B 100d	A	85.13%	85.15%
Bidirectional SimpleRNN	GloVe 6B 100d	C	70.31%	68.22%
GRU	GloVe 6B 100d	A	87.80%	87.72%
LSTM	GloVe 6B 100d	B	85.35%	85.18%
Logistic Regression	TF-IDF	LR_3	91.20%	91.05%
Multinomial NB	TF-IDF	NB_1	84.96%	84.57%
Random Forest	TF-IDF	RF_1	90.20%	90.17%
SimpleRNN	GloVe 6B 100d	B	49.19%	45.10%

Table 4: Selected configurations after validation.

Rank	Model	Representation	Accuracy	Macro-F1	Weighted-F1	Time (s)
1	BERT Base	BERT WordPiece	95.16%	95.06%	95.16%	1235.44
2	Bidirectional GRU	GloVe 6B 100d	90.59%	90.56%	90.64%	46.58
3	Logistic Regression	TF-IDF	90.59%	90.27%	90.56%	9.66
4	Random Forest	TF-IDF	89.64%	89.58%	89.61%	10.79
5	GRU	GloVe 6B 100d	87.64%	87.56%	87.69%	34.42
6	Bidirectional LSTM	GloVe 6B 100d	85.25%	85.14%	85.36%	51.68
7	LSTM	GloVe 6B 100d	84.86%	84.76%	84.92%	36.47
8	Multinomial NB	TF-IDF	84.24%	83.58%	84.09%	0.04
9	Bidirectional SimpleRNN	GloVe 6B 100d	73.00%	71.36%	72.58%	76.47
10	SimpleRNN	GloVe 6B 100d	48.44%	44.38%	46.56%	28.26

Table 5: Final test performance. Time is the recorded tuning-run time in the reported environment.

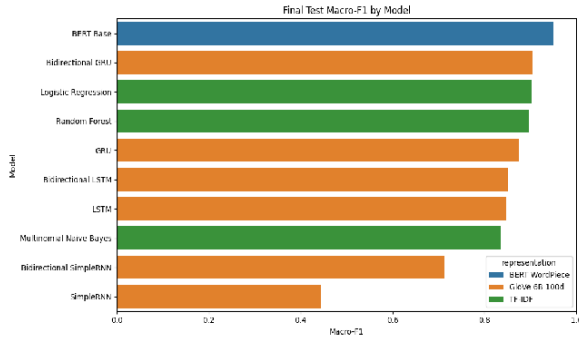


Figure 4: Final test macro-F1 scores for all ten models.

4.3 Traditional Model Comparison

Logistic Regression is the best classifier in the three TF-IDF models. The final macro-F1 is 90.27% for it, while Random Forest and Multinomial Naive Bayes end at 89.58% and 83.58%, respectively.

This result implies that this dataset has a rich and sparse lexical representation. There is a special terminology associated with scientific domains, and therefore, linear decision boundaries over TF-IDF features can be used to separate many classes. Another critical computational advantage of Logistic Regression over BERT is the recorded tuning-run time (about 9.66 seconds) versus BERT’s recorded tuning-run time of 1235.44 seconds in the reported environment.

Random Forest has a slightly lower accuracy than Logistic Regression. This can suggest that the high-dimensional sparse TF-IDF space is more appropriate for a linear classifier than tree-based partitioning.

The traditional-model performance is lowest for Multinomial Naive Bayes. It is very rapid but it is limited by some conditional-independence conditions for scientific abstracts with related terms and multiword concepts.

Thus, the answer to RQ1 is Logistic Regression, which has the best macro-F1 among the traditional classifiers.

4.4 Recurrent Model Comparison

This can be observed from the recurrent experiments, which confirm that the gated recurrent architectures are better than the simple SimpleRNN. Bidirectional GRU outperforms all other recurrent models (macro-F1 score of 90.56%).

The GRU gating mechanism achieves a significant improvement over SimpleRNN as shown by the GRU without bidirectionality, which achieves 87.56%. Another advantage of Bidirectional GRU is its ability to learn from both directions, which helps to enhance its performance.

The accuracy of LSTM is 84.86% and that of Bidirectional LSTM is 85.25%. In this experiment,

complexity of the architecture in itself does not guarantee a higher performance. The GRU-based architecture outperforms LSTM significantly.

SimpleRNN is the smallest recurrent model, with a macro-F1 score of merely 44.38% on the test set. The bidirectional SimpleRNN model outperforms it with 71.36%, with the information from both directions helping significantly but not quite to the level of the gated recurrent models.

Hence, Bidirectional GRU is the answer to RQ2. Recurrent cells tend to improve recurrent performance, although the type of recurrent cells is still important. GRU achieves the best sequence modeling ability to computational cost ratio in these experiments.

4.5 BERT vs. the Non-BERT Models

BERT improves macro-F1 by approximately 4.50 percentage points over Bidirectional GRU, the best non-BERT model. However, its recorded tuning-run time is substantially higher. Logistic Regression achieves 90.27% macro-F1 with a much shorter recorded runtime, making it a strong alternative when maximum predictive performance is not required.

Therefore, RQ3 cannot be answered using predictive performance alone. BERT provides the highest macro-F1, while Logistic Regression provides the strongest observed performance–runtime trade-off. A complete cost-effectiveness evaluation would also require inference latency, memory usage, hardware utilization, and monetary cost.

4.6 Confusion Matrix and Error Analysis

All ten models were used to create confusion matrices, both for raw and row-normalized representations.

The most common BERT error is Biochemistry being predicted as Medical Science, occurring 11 times. The second most common confusion is Mechanical Engineering being predicted as Civil Engineering, occurring 8 times. Electrical Engineering is predicted as Mechanical Engineering 7 times.

Other notable errors include seven Psychology abstracts predicted as Medical Science and seven Medical Science abstracts predicted as Biochemistry.

These errors are valid because there is some semantic overlap in the meaning of the categories that are confused. Biochemistry and Medical Science often talk about biological processes, disease, proteins, mechanisms and experiments in the lab.

True Domain	Predicted Domain	Errors
Biochemistry	Medical Science	11
Mechanical Engineering	Civil Engineering	8
Electrical Engineering	Mechanical Engineering	7
Psychology	Medical Science	7
Medical Science	Biochemistry	7
Medical Science	Psychology	6
Electrical Engineering	Civil Engineering	5
Civil Engineering	Mechanical Engineering	4
Mechanical Engineering	Computer Science	3
Computer Science	Mechanical Engineering	3

Table 6: Most frequent off-diagonal errors for BERT.

Likewise, the terms of Civil Engineering and Mechanical Engineering can share the terms associated with structures, materials, thermal systems, energy, and experimental engineering.

The errors thus illustrate a noteworthy aspect of the task, that of some abstracts being truly interdisciplinary or having terms that are common to several parent domains. The data set itself assigns any document to one category of parent, regardless of the content of the scientific information.

This directly answers RQ4. The most critical confusion is Biochemistry–Medical Science and next the engineering domain confusions. These errors may be related to the interdisciplinary studies and to the scientific vocabulary that is shared.

5 Conclusion

In this study, we tested automatic scientific-domain classification with ten models on three representation families: TF-IDF, recurrent representations based on word vectors from GloVe and BERT WordPiece representations. The experiments were run on 11,967 scientific abstracts belonging to seven parent domains that were drawn from the WOS-11967 dataset.

The results show that contextual transformer representations are the best for classification. The test accuracy of BERT reaches 95.16% and macro-F1 reaches 95.06%, which is the best model in the experiment. Bidirectional GRU is the best recurrent and best non-BERT model with macro-F1 of 90.56%. Logistic Regression is the best traditional baseline model, achieving 90.27% macro-F1.

Additionally, the recurrent experiments show that model architecture plays a role. Long short-term memory (LSTM) networks are not as effective

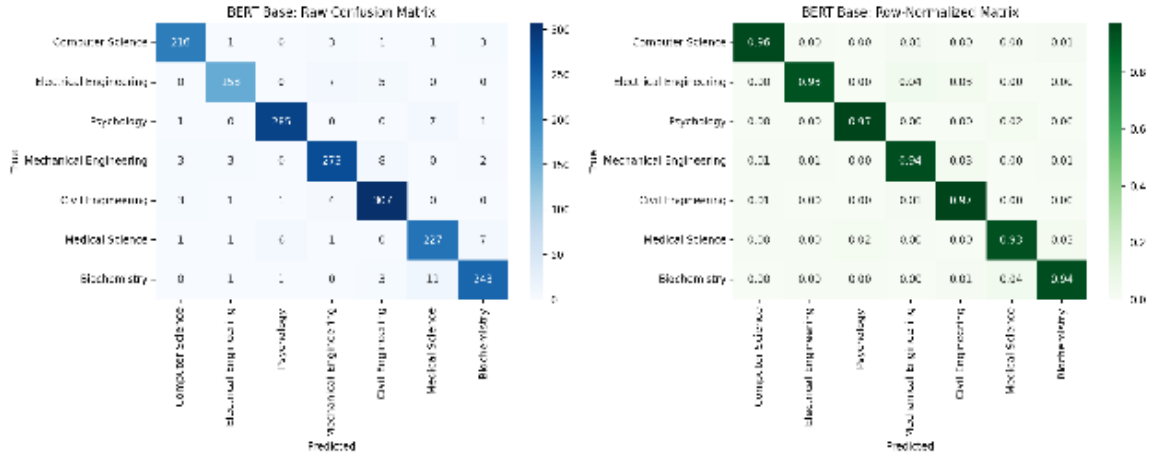


Figure 5: Raw and row-normalized confusion matrices for the best-performing BERT model.

as gated recurrent unit (GRU) networks, and GRU offers good performance along with a moderate amount of computation. Bidirectional GRU works best, indicating that context on both sides of an abstract is beneficial to the scientific-domain classification task.

The traditional experiments show that TF-IDF is still a very good basic system. Logistic Regression achieves nearly 90% macro-F1 and needs just a tiny bit of the training time of BERT. This makes it a good option when computational resources and/or response time are significant.

Some future research directions include using domain-specific pretrained embeddings, scientific-language transformer models such as SciBERT, multi-level classification with parent and child labels, extending the context window of the transformer models, and ensemble methods combining lexical and contextual representations. A second possible track to explore is multi-label classification, that is, papers in different scientific fields can have more than one label and not be limited to one.

Overall the experiments demonstrate that modern contextual representations offer significant benefits for scientific-domain classification, while simpler representations are competitive, and can offer a very good accuracy-to-computation trade-off. The results indicate that it is not enough to evaluate predictive accuracy when selecting a model; other factors include application needs, computational resources, and the interdisciplinary, scientific literature.

Limitations

There are a few constraints to be taken into account. First, it has seven parent categories and

interdisciplinary papers still have to be assigned to one of these. Second, the class frequencies are not equal; this is why macro-F1 is used as the primary selection criterion, as opposed to accuracy. Third, GloVe is a general-purpose embedding and might not be as good as domain-specific embeddings for conveying specialized scientific terms. Fourth, the six recurrent models and BERT require fixed maximum sequence lengths selected from the training-length distributions. Lastly, the training time comparison recorded is specific to the Kaggle hardware and software environment and should not be considered as a standard.

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